

Expert Evaluation of Haptic Virtual Reality User Interfaces for Medical Landmarking

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ABSTRACT

Three-dimensional (3D) models have widely been used in medical diagnosis and planning tasks. Haptic virtual reality (VR) interfaces implemented by using VR equipment and haptic devices have previously been proposed for these medical 3D manipulation tasks. They have been found to be faster and more accurate in a medical marking task with novel users, compared with the traditional 2D interaction technique that uses a mouse and a 2D display. In this study, we recruited medical experts who do the medical landmarking task as part of their daily work to examine the performance of haptic VR interfaces and to investigate experts' user experience. There were no statistically significant differences between the haptic user interfaces and the mouse-based 2D interface in terms of task completion time and marking accuracy. Based on experts' subjective data, haptic VR interfaces showed great potential for medical work because of the natural input methods and haptic feedback.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; **Virtual reality**; *User studies*.

KEYWORDS

3D visualization, haptic feedback, force-feedback device, model marking

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1 INTRODUCTION

Computer-aided medical diagnosis and planning based on medical digital imaging is an important task for radiologists [17]. For example, in cephalometry analysis, radiologists highlight critical points on the images of the human skull as points of reference (i.e.,

anatomical landmarking [10]), and the accuracy of landmarks directly associates with measurements in the cephalometry analysis and thus influences the overall quality of medical diagnosis and planning [12]. Identifying and marking cephalometric landmarks on 2D images is a challenge because of the overlapping and duplicated skull structures (e.g., [4, 12]). As an alternative, 3D cephalometry analysis has been proposed [9, 14]. The volumetric medical images help radiologists better understand the spatial structure of the skull and the locations of landmarks, potentially improving the precision of medical landmarking. However, efficient 3D cephalometry analysis is still considered to be difficult and time-consuming [18, 21].

The user interface plays a critical role in user performance for these medical 3D manipulation tasks. To interact with volumetric anatomical images, current systems mainly utilize the traditional 2D interaction technique of a 2D display as the visual output channel and a mouse as the input tool. Previous studies (e.g., [2, 5, 8]) have argued that using 2D interaction devices such as the mouse for 3D manipulation (i.e., rotate, pan and zoom) can be difficult. For example, Bade *et al.* [2] demonstrated the issues of the existing mouse-based rotation techniques, such as Virtual Sphere [6] and ArcBall [15], for 3D manipulation. Multiple studies have examined using tangible interfaces (e.g., [3, 8]) or touchscreen-based user interfaces (e.g., [23]) for 3D manipulation tasks. However, the touchscreen-based interface could not exceed the mouse in terms of interaction efficiency (e.g., [23]). Tangible interfaces require additional input devices for other operations on the 3D model (e.g., [3]), such as marking and measuring, and application has been limited.

To address this issue, two haptic VR user interfaces have been proposed for medical 3D manipulation, implemented by using VR headsets, controllers, and haptic devices [11] (see Figure 1). These VR interfaces could enable intuitive and realistic 3D interaction (a flexible head movement-based viewing perspective, natural hand-based input and haptic feedback together), thus promising for the tasks involving 3D manipulation [5], like the medical landmarking task. Through comparing with the mouse-based interface as the baseline, these interfaces showed advantage, in terms of interaction efficiency, marking accuracy and user experience with novel users [11].

To explore the potential of haptic VR interfaces to be adopted by professionals, we invited a group of medical experts to conduct a comparative study. Their background and years of work experience could also provide a new perspective to evaluate two haptic VR interfaces. Six radiologists were recruited from the local hospital community as the experimental participants. We used the system developed by Li *et al.* [11] with the permission of the authors. Two

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Figure 1: A head-mounted VR display and either a VR controller implementing the vibrotactile VR interface (left) or a force-feedback device implementing the kinesthetic VR interface (middle) were used for medical 3D manipulation tasks [11]. An example of the skull model (right) with selected cephalometric landmarking points (red dots).

haptic VR interfaces were evaluated based on a Steiner cephalometric landmarking task [13]. As in the previous study [11], the mouse-based interface (2D display and mouse) was used as the baseline in the experiment, because it is the current popular and dominant interface in the medical field. To explore the practical usability of the two VR interfaces, it is necessary to experimentally compare the two haptic VR interfaces with the traditional mouse-based interface. To evaluate the two haptic VR interfaces, we collected both objective and rating-based subjective data. The objective data included task completion time and marking accuracy. The subjective data included the perceived mental effort, hand fatigue, naturalness, immersiveness and user preference. We also collected the participants' comments and opinions for the two VR interfaces to reveal insights into the subjective results. These data could help us further understand the strengths and weaknesses of current haptic VR interfaces and make an empirical basis for their future development.

The paper is organized as follows: First we introduce the methodology (Section 2) and the experiment (Section 3) including the apparatus, participants, and the procedure. Then we go through the results (Section 4). In the end, we discuss the experiment results (Section 5) and conclude with the main findings of the study (Section 6).

2 METHOD

2.1 Design of the experimental system

We used a prototype marking system described by Li *et al.* [11]: The system included three different user interfaces: vibrotactile VR interface using a handheld controller ("VR+Controller"), kinesthetic VR interface using a force-feedback device ("VR+Pen") and traditional mouse-based 2D interface ("Mouse+Display"). Both haptic VR interfaces utilized a VR headset ([22]) as the display. There would be vibrotactile feedback (vibrotactile VR interface) or force feedback (kinesthetic VR interface) while touching the model surface. The baseline mouse-based interface used a standard 2D display and a mouse without haptic feedback.

2.2 Experiment design

We used a within-subject experiment design and the experimental task was the medical landmarking on 3D models. Three patients'

skull models were used in the experiment. The permission to access these models for research was granted by the local hospital. For each user interface, participants were asked to do three marking tasks and each task involved one skull model. For each model, six out of 11 Steiner cephalometric landmarks [13] were randomly selected as marking positions visible as red dots. Thus, participants were required to mark 18 positions (3 models X 6 positions) while using each interface. They employed the input tools to manipulate the model for searching the marking positions, and then reach and touch (using the controller or the force-feedback device) or point at (using the mouse) the positions to determine where they wanted to mark. The marking method using the mouse-based interface was based on the ray-casting technique [7]. Markers were created by pressing the space key on the keyboard. The order of marking the six positions could be freely decided by the participants.

2.2.1 Objective measures. We measured the task completion times and the marking accuracy. The task completion times were measured from the time of onset of the task (time of showing the model) to the time of marking of the last, the sixth position. The marking accuracy was measured as an Euclidean distance between the target position and the marker position. For each task we collected one completion time and six marking accuracy figures.

2.2.2 Subjective questions. We asked the participants to answer a few subjective questions. In the first set of questions, "Evaluation of the interaction method" (see Table 1), answered after each condition, we asked how the participants felt about several aspects of the experiment. We used a 7 step Likert scale for the answers, from 1 (Not-at-all) to 4 (Somewhat) to 7 (Very). For questions S4, S6 and S7 we asked the reasons for their answers.

In the second set of questions, "Ranking of the interaction methods" (see Table 2), answered at the end, we evaluated how the methods compared to each other by asking the participants to provide a ranking of the methods (by marking them as 1st, 2nd and 3rd). For all questions we also asked why that specific ranking.

2.3 Pilot study

Four participants who had experience using the VR equipment and the force-feedback device were recruited to conduct a pilot study to determine the system parameters. The sizes of skull models were original without any scaling. The size of interaction points for the

Table 1: The questions S1 to S7 to evaluate the subjective impressions and asked immediately after each interaction condition.

S1	How confident you were in your ability to use the interaction method?
S2	How easy/effortless was the interaction method to use?
S3	How tired are your hands?
S4	How natural was it to use?
S5	How immersive was the task environment?
S6	How easy was it to orient yourself in the environment?
S7	Could you imagine using the method daily?

controller and force-feedback device (0.4 cm diameter) and the size of marking points (0.4 cm diameter) were selected, so that they were clearly visible to the user. Simultaneously, these sizes were small and equal to minimize their influence on marking accuracy. The force-feedback device had no movement scaling, and thus the user's hand movement matched with the movement of the interaction point to enable good control of the interaction point. The rotation factor (3.5) and zooming factor (6) for the mouse-based interface were selected to allow the users to use the mouse to do the landmarking task without any operational difficulty. The parameters for vibrotactile feedback of the VR controller (80 ms duration) and force feedback of the force-feedback device (stiffness coefficient 1 and friction coefficient 0.7) were also selected. These parameter choices could allow the user to clearly perceive the haptic feedback while touching the skull models.

3 EXPERIMENT

3.1 Apparatus and participants

We used an HTC Vive VR headset [22], with a basic Vive controller and Geomagic Touch X force-feedback device [1] as input devices for the two VR conditions, respectively. For the mouse-based 2D interface, we used a Samsung 245B Plus 24" LCD display with a computer mouse. A standard keyboard was used to create the markers. The experimental system was developed using Unity system [19], along with SteamVR [16] and Geomagic OpenHaptics plugin [20].

The participants were radiologists recruited from the local hospital. They had a long experience of the 2D display and the mouse-based interaction method, but no experience of the other methods with the VR headsets and haptic devices. To compensate for that we offered the participants sessions where they could practice using

Table 2: The questions R1 to R3 to rank the interaction methods and asked after all the interaction conditions were tried out.

R1	Which interaction method did you like the most?
R2	Which interaction method did you think would be the easiest to adopt by a novice?
R3	Which interaction method had the most potential if it could be developed further?

the interaction methods in other 3D manipulation tasks, not in the medical marking task. There were six participants and their age varied from 25 to 60 years; the mean age was 39 years. Two of the six participants were female and four were male. The participants' experience on using medical marking tools varied from 2 to 36 years, the mean value being 14 years.

3.2 Procedure

Upon arrival, each participant was introduced to the study and signed the informed consent. Questions were asked about background information. The order of conditions was balanced between the participants using Latin squares. For each condition, the participant could be practicing up to 10 mins to be confident enough for the control of the examined interface (particularly the VR interfaces). When the participant was ready the data collection was started. The participants were asked to do the experimental task as accurate and fast as possible.

For each condition there were three tasks, each task having six marking positions. In the beginning of each task the participant saw a skull model with six marking positions, where some positions were on surfaces behind the model and not immediately visible. The participant was to pick up the model and look for a marking position and make a marking, repeating that for six times. For each marking position, the participant had only one chance to mark. After completing a task, the participant would automatically be presented the next task until all three were done.

After each condition the participant filled in a condition evaluation form, and then started a next condition, until all three were done. After going through all conditions, the participant filled in a condition ranking form.

4 RESULTS

4.1 Task completion times

Each participant completed three conditions, and three marking tasks for each condition. That is, we recorded nine task completion times. To compare the completion times between conditions, we first calculated the median values of completion times for each condition per person. Then, the median value for each condition over participants and the median of absolute deviation (MAD) have been calculated. The results are listed in Table 3.

Kruskal-Wallis H-test indicated statistically significant differences among three user interfaces (p -value = 0.002), but pair-wise comparisons didn't show any. However, by observing the data, the vibrotactile VR interface and the mouse-based interface performed fast but using the kinesthetic VR interface led to longer task completion time.

Table 3: The median and MAD values of task completion times.

Condition	Completion times	MAD values
Mouse+Display	39 s	5 s
VR+Controller	46 s	7 s
VR+Pen	96 s	19 s

4.2 Marking accuracy

Each participant completed nine marking tasks (three tasks $\times$ three conditions) with six separate marking positions in each task. To compare the marking accuracy between conditions, we first calculated the median values of accuracy for each task separately, and then the median values for each condition per person, and finally the median value and the median of absolute deviation (MAD) over participants. The results are listed in Table 4. There were no statistically significant differences among three user interfaces in the Kruskal-Wallis H-test.

Table 4: The median and MAD values of marking accuracy.

Condition	Marking accuracy (cm)	MAD (cm)
Mouse+Display	0.300	0.053
VR+Controller	0.374	0.050
VR+Pen	0.380	0.034

4.3 Subjective data

4.3.1 Evaluations of the interaction methods. The interaction methods were evaluated using seven questions (see Table 1). We asked the participants to give also a free-form reasoning of their evaluations for three of the questions (S4, S6 and S7 in Table 1) to get a better understanding of their views of the interaction methods. The evaluation values are shown on Figure 2. There were no statistically significant differences among three user interfaces in the Kruskal-Wallis H-test for any of the questions.

For the question of how natural the given interaction method was to use (S4), the answers mostly referred to familiarity. Five out of six participants commented that mouse is natural as it is so familiar. There were several comments of the controller feeling natural because of strong feeling of 3D and good responsiveness. The only negative comment was of hand fatigue. The force feedback device was commented to feel natural because of easy control of the moves and the realism of the model surface. It got negative comments about the model surface because of technical difficulties and the restricted movement of the device.

For the questions of how easy it was to orient oneself (S6) and using the technique daily (S7), the answers were similar. The mouse was familiar and felt easy to orient, and used daily, already. The controller received positive comments about the VR environment with 3D depth perception and ease of use and orient. It could be used daily but was considered too big. Regarding the force-feedback device, the participants generally commented the force feedback was useful. Some participants commented that the device was felt easy to use but some stated not to use it daily, as it was difficult.

4.3.2 Ranking of the conditions. The interaction methods were ranked in three dimensions (see Table 2). The ranking results are listed in Tables 5 and 6. The participant comments were separately analyzed to identify common items and to find possible ways of improvements.

Half of the participants ranked the controller the most liked, it felt natural and easy to use, and the VR environment provided the

Table 5: The ranking results of the interaction methods as the most liked, the second and the least liked.

Condition	1 st	Ranking 2 nd	3 rd
Mouse+Display	2	3	1
VR+Controller	3	3	0
VR+Pen	1	0	5

3D perception. Mouse interaction was also liked as familiar and easy to handle. One participant liked the force feedback device the most, the interaction method felt natural and pen-form device was nice to use.

In the analysis we noticed that the rankings by easiest to adopt were identical to the rankings by liking with similar comments. Controller was felt natural and easy to use, mouse was ergonomic and had a great accuracy, and the force feedback device had a realistic feel of objects.

Table 6: The ranking results of the interaction methods as having the most development potential, the second and the least potential.

Condition	1 st	Ranking 2 nd	3 rd
Mouse+Display	0	0	6
VR+Controller	3	3	0
VR+Pen	4	2	0

Four participants out of six ranked the force feedback device as having the most future potential, while three out of six ranked the controller as most promising. (One participant ranked both as 1st.) The positive comments were about the virtual reality and realistic feel of structures. The VR display should be improved. The mouse was ranked having the least potential as it cannot be developed further.

4.3.3 General comments. Haptic feedback was felt helpful, the force feedback device being even more realistic than the vibrotactile feedback of the controller. The participants raised questions about the ergonomics of the VR display as it is somewhat heavy and might affect eyes if used for longer times. The display resolution of the VR device was felt as low and the controller was considered heavy for use. The VR environment with a realistic 3D feel was approved.

5 DISCUSSION

5.1 Analysis of objective data

Regarding task completion times, although there were no significant differences among three interfaces, the kinesthetic VR interface using a force-feedback device had a clearly lower task performance than other interfaces, shown in Table 3. The participants were familiar with the used Steiner cephalometric landmarks [13] in their daily work, and thus could effortlessly find the landmarks without need for complex manipulation of the models (i.e., rotate, pan and zoom). This result was similar to the previous findings

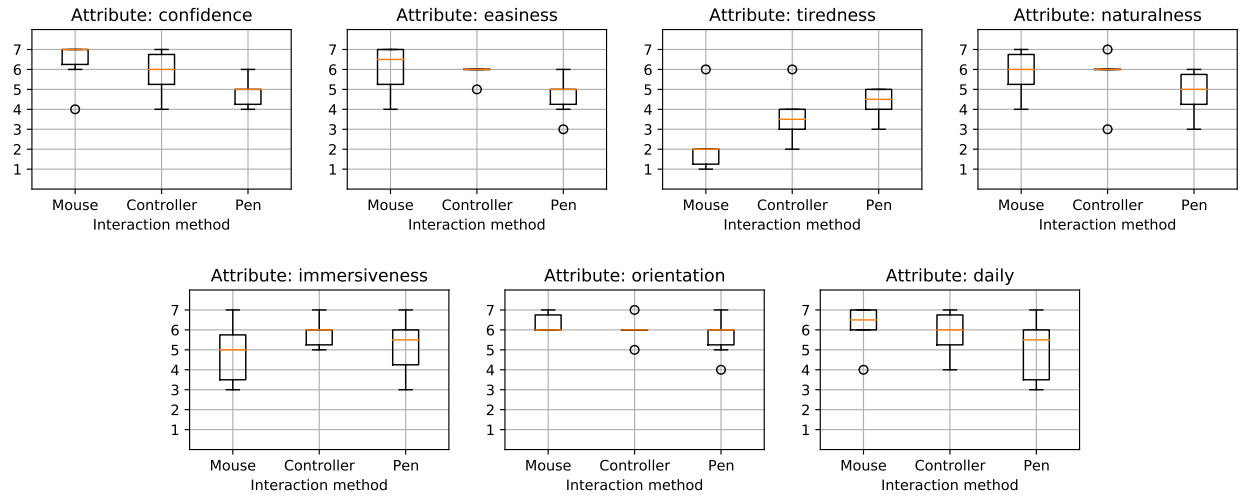


Figure 2: The subjective evaluation values of each interaction method for each asked attribute (see Table 1).

(i.e., the easy-to-mark tasks in [11]) that both the controller-based and the mouse-based interfaces could achieve fast interactions for easy 3D manipulation tasks and the limited hand motions using the force-feedback device might negatively affect the interaction speed.

Regarding the marking accuracy, the previous study demonstrated that the kinesthetic VR interface with force feedback could have higher marking accuracy than vibrotactile VR interface and the 2D interface [11]. This was because, unlike the vibration feedback which could only be used as a simple signal for surface contact, force feedback could help accurately locate the target position and stabilize the hand while marking. However, our results showed that there were no statistically significant differences among the three user interfaces. The medical professionals likely have developed effective strategies for the landmarking task using the mouse-based interface following their long medical experience, which thus largely improved the task performance of the mouse-based 2D interface in terms of marking accuracy.

Overall, for the medical professionals who have long previous experience with the landmarking task using the 2D interface, there were no significant differences among three user interfaces in terms of task completion time and marking accuracy. This may be partly because there were only six participants in the study (due to the limited number of radiologists available in the local hospital). However, by the observation of data, it is clear that using two VR interfaces generally had lower performance than using the 2D interface. This is understandable because of the participants' years of work experience using the 2D interface. However, haptic VR interfaces are still promising for medical diagnosis and planning as the user performance is expected to get improved with more practice. A long-term study with more professional participants is required to verify this statement. In addition, we used the 2D interface (2D display and mouse) as the baseline in the study, because this interface is still dominant in the medical field. The present study demonstrated

the strengths and weaknesses of the two haptic VR interfaces and their potential to be adopted by medical professionals by comparing them with the 2D interface. Combining the mouse and the VR headset as another user interface is technically possible. However, it is not clear that whether the mouse would be an efficient input device for an interactive virtual environment provided by the VR headset. Because of that we did not use the interface that combines the mouse and the VR headset as the experimental baseline. This could be examined in the future studies.

5.2 Analysis of subjective data

As there were no significant differences in the objective measures, it is interesting to study the expert participants' subjective feelings and comments. It is instructive, for example, how the experts compared the haptic VR interaction methods against the current well-established mouse-based interaction method and what kind of improvements might be expected from using the proposed methods.

Based on the observations from subjective data, the mouse-based interface was overall perceived easier than the haptic VR interface although there were no significant differences. That is understandable as probably related to the long experience by these experts on the mouse-based interaction in similar medical tasks. In contrast, they only had a short time to familiarize themselves with the two VR interfaces. However, when the participants ranked the interaction methods based on personal preference and future potential, the results gave a very different outcome (Tables 5 and 6). Four out of six participants reported that the haptic VR interfaces were their most preferred interaction methods and easy to adopt by them, and all participants reported that the haptic VR interfaces have the largest potential for medical work in the future.

Based on the ranking data of personal preference (Table 5), three participants liked the controller-based interface the most as the controller was felt natural and easy to use. One participant liked the kinesthetic VR interface the most since the pen-like end-effector

of the force-feedback device was natural to keep and nice to use. All the participants emphasized that the VR environment enabled by the VR head-mounted display provided better 3D perception. This is encouraging and demonstrates the strong effect that the 3D environment has on the use. Even as the users were rather unfamiliar with the VR headsets, the handheld controller, and the force-feedback device, both haptic VR interactive systems had been reported to be potential to adopt by the medical professionals.

Based on the participants' comments for their personal preferences, while the vibrotactile VR interface was found natural and easy to use due to the completely free hand movement as the input, the controller itself should be made much smaller and lighter, as the current implementation was considered heavy and thus tiring for longer time usage. For the kinesthetic VR interface, although the device also allowed free hand movement, the hand gestures that the user could use were not as flexible as the ones when using the controller. For novel users who were less familiar with the mechanical structure and workspace of the device, this might lead to the operational difficulty and have negatively influenced their subjective feelings. It could explain why the VR interface that used a force-feedback device was considered as the most difficult interface to adopt (five out of the six participants). However, we believe that this situation could be improved with more use experience.

Although there were issues related to haptic VR interfaces, all participants considered the haptic VR interfaces as having more future potential than the mouse-based interface (Table 6). Four out of six participants considered the VR interface that used a force-feedback device as being the most potential interface and three participants considered that the controller-based interface was the most potential one (one participant selected both). No participants considered the mouse-based interface as having the most future potential when comparing it with two VR interfaces.

All participants mentioned that haptic feedback, especially kinesthetic feedback, was helpful for their work. Other reasons mentioned included the free hand-based interaction method and the realistic 3D environment provided by the VR headset. The display resolution of the VR headset was commented to be low but no participant mentioned that as affecting the landmarking tasks. According to the post-test interviews with the participants, they have noticed that the mouse-based interface based on the ray-casting technique [7] could easily make inaccurate and erroneous marking points following their long-term work experience. Using hand motions to manipulate and mark the volumetric medical images with haptic feedback as the confirmation of surface contact was commented as an effective interaction method to address this issue.

6 CONCLUSION

We recruited medical experts to evaluate two haptic VR interfaces by comparing them with the traditional 2D interface used by the experts. The objective measures showed that there were no statistically significant differences among the three user interfaces for these experienced users. Based on the subjective responses and comments, two haptic VR interfaces have their own shortcomings at the current stage. However, all medical experts agreed that the two haptic VR interfaces have great potential to be used in future medical workflow.

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